PROCESS OR MATERIALS REVIEW REQUEST FORM

SNF Stanford Nanofabrication Facility

Request Title: **Processing ScAlN in AZ-400K and Microposit MF-319**

Requestor: Payton Broaddus **Date: 2021/03/08**

Non-SNF staff, please complete the following:

Badger ID: Broaddus

Company name/PI: solgaard

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material.** Please provide all common names, trade names, and CAS numbers where appropriate. Include an MSDS, if available; or provide the reason, if not. Make sure to include information for any new secondary chemicals (such as a developer for a new resist). Read the MSDSs as well as the Stanford Chemical Storage Groups to determine the Storage Group Identifier and Main Hazard Class of your chemical/material.

Chip: 500nm ScAlN on 100nm Mo on 500um Si

AZ-400K: Potassium containing Developer

Hazardous ingredients:

Component	CAS-no. (Trade secret no.)	Concentration
Potassium borates	20786-60-1	< 15 %

Non-hazardous ingredients:

Component	CAS-no. (Trade secret no.)	Concentration
Water	7732-18-5	> 85 %

Microposit MF-319: TMAH containing Developer

Water 7732-18-5 > 95.0 %
Surfactant < 1.0 %
Tetramethylammonium hydroxide 75-59-2 2.2 %

2. **Vendor/manufacturer info:** address and phone number, website URL.

AZ-400K:

https://www.microchemicals.com/products/developers/azr_400k_developer.html
Microchemicals GmbH | Nicolaus-Otto-Str. 39 | 89079 Ulm | Tel.: +49 (0) 731 977 343-0 | E-Mail: info@microchemicals.com

Microposit MF-319:

Rohm and Haas Electronic Materials LLC 455 Forest Street Marlborough, MA 01752 United States of America

3. **Reason for request:** Please give serious thought to this. If you have any process information (application notes from the vendor, protocol from another lab, experimental methods section of an article), please include it, preferably as attachments. Ask yourself these questions: Is this the latest procedure? Are there newer/safer alternatives that will also work for my project? Will any of the current SNF approved chemicals and materials work for me?

The etch speed of AZ-400K and MF-319 is related to temperature. We need more than A/min etch rates so we are requesting 60C for faster etch rates as described in other papers. According to SEM data the etch rate with 60C AZ-400K is 5nm/min +- 2nm/min when using the hot plate method. This process requires better control over temperature to reduce error bar on etch rate, which will require the use of a controlled temperature bath.

We are actually using Al etchant for the first etch step as a bulk etch of the ScAlN, but because of the high selectivity towards underlying Mo we cannot use Al etchant to clear the Mo pads. Furthermore both AZ-400K and MF-319 are incompatible with PR masks, so we cannot use these for the bulk etch. Also, as discussed at the SNF process clinic we cannot use Dry etchers with ScAlN due to the low reactivity of the Sc radicals.

4. **Process Flow:** Please provide a detailed process flow description on how and where you proposed to use this chemical. This should include **all Lab equipment** to be used for processing your wafers once your new chemical or material has been used (even if your new material is a film that is removed, it may still pose potential contamination concerns.) Make sure to include wet benches. Please note that f the chemical/material is to be used in any the "clean" equipment, purity specifications will be needed. This is most important for chemicals/material that are not normally used for VLSI device fabrication. To be allowed into a "clean" tool, the material should MOS grade or better.

Starting with: 1mmx1mm ScAlN chip

1. Enabled wbflexcorr-3/4 and wbflexsolv-1
2. Moving forward all movements of any beaker and chemicals or teflon containers will be in slow and deliberate motions to reduce chance of cracking beakers and reducing agitation of chemicals during chemical etch (Recommendation from Uli)
3. Prepare Acetone, Methanol, Isopropyl and DI water beakers at wbflexsolv-1 and label them with chemical cards
4. Submerge chip 1 min in Acetone, Methanol, then Isopropyl
5. Submerge chip in DI water and safely transport to wbflexsolv-1
6. Dispose of Acetone Methanol and Isopropyl and clean glassware
7. Disable wbflexsolv
8. In wbflexcorr wash 5 beakers with DI water and N2 gun clean
9. Fill 3 beakers with DI water and label them with chemical cards
10. Make sure constant temperature bath (CTB) has enough water
11. Set CTB to 60C and wait 2 hours for temp to stabilize
12. Put on chemical suit
13. Transport AZ-400K using cart
14. Pour AZ-400K into one beaker and MF-319 into another beaker
15. Return AZ-400K and MF-319 to back room storage using cart
16. Stay at station for remainder of process
17. Place thermometer into both beakers
18. Place single beaker in constant temperature bath (CTB)
19. Wait 30 minutes for temperature to stabilize
20. Place chip in teflon chip holder with teflon coated tweezers
21. Lower teflon chip holder into beaker
22. Remove teflon chip holder after N minutes (40 minutes max)

23. Dunk teflon chip holder into 3 separate DI water beakers
24. N2 dry chip with teflon coated tweezers
25. Turn off CTB
26. Wait until thermometer says the solutions are room temperature
27. Dispose room temperature AZ-400K and MF-319 into separate labeled empty containers using funnel
28. Clean glassware with DI water and use funnel to dispose of potentially contaminated DI water into container
29. Place fulled filled out containers in pickup station
30. Disable wbflexcorr
31. Remove chemical suit

5. **Amount and form.** How much will you bring in? Is it solid, powder or liquid? (Note: as a general rule, powders are not permitted in the cleanroom.) Do you need to mix it to use it?

AZ-400K and Microposit MF-319 are already in the lab. Liquid. Less than 100mL poured into beakers. No mixing required.

6. **Storage:** Will you be storing your chemical/material at SNF? If so, please note any potential reactivity's (this should be on the MSDS). Storage groups A, B, D and L are stored in the yellow solvent cabinet in the furnace support area, while storage groups C, E, F and G are stored on top of one of the Pass-through Carts. Ensure your chemical container or material is properly labeled. If there is no available room, it must be stored by in the bulk storage area. You will then need to obtain it from receiving area personnel each time you want to use it and return it to them when you are finished using it (or each time you leave the lab). Note that there is no storage of chemicals/materials in the processing lab or at any wet bench.

AZ-400K and MF-319 are already stored in SNF. No changes to storage procedure.

7. **Disposal**: How will you dispose of any waste or excess chemical or material? In your discussions with experts and vendors, try to determine the best way to dispose of your spent chemicals and by-products. Please refer to the SNF Lab members Safety Manual for the different methods of waste disposal that are available in the lab.

Standard disposal. Fully label empty container and all waste goes into it. Container placed in pickup station.

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: